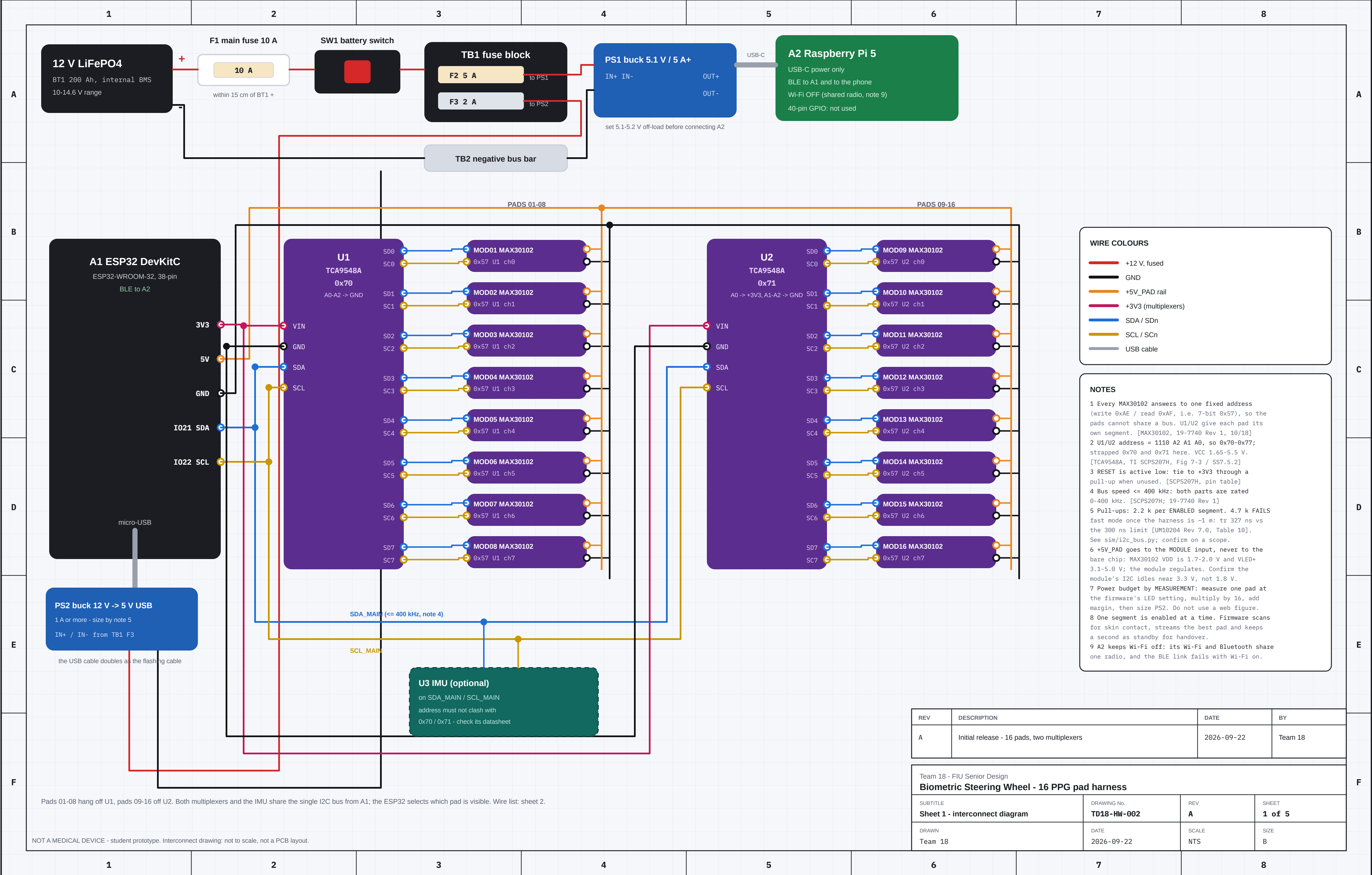




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B	DEVICE DATA - AS VERIFIED AGAINST THE PRIMARY DATASHEETS TCA9548A TI SCPS207H, MAY 2012 - REVISED SEPTEMBER 2024 8 bidirectional channels; address byte 1110 A2 A1 A0 -> 0x70-0x77 (Figure 7-3, section 7.5.2) VCC 1.65-5.5 V (recommended operating conditions); fSCL 0-400 kHz; 5 V tolerant I/O RESET active low; connect to VCC through a pull-up when unused (pin functions table) MAX30102 Maxim 19-7740; Rev 1; 10/18 I2C write address 0xAE, read address 0xAF (7-bit 0x57); fSCL 0-400 kHz (electrical characteristics) VDD 1.7-2.0 V (1.8 typ); VLED+ 3.1-5.0 V (3.3 typ) - the module's regulator makes 5 V VIN safe FIFO depth 32 samples - read in bursts instead of switching channel per sample IMU U3 address unverified here: confirm against the datasheet of the module actually fitted, and check it does not clash with 0x70 / 0x71								B																																																																																												
	C	BEFORE POWER-UP 1 Set PS1 to 5.1-5.2 V with no load, then connect A2. 2 Check +5V_PAD and +3V3 at the far end of the harness, not only at the source. 3 With power on and no I2C traffic, SDA and SCL must idle high near 3.3 V. 4 Scan the bus: only 0x70 and 0x71 answer until a channel is enabled. 5 Enable one channel at a time and confirm exactly one 0x57 answers. 6 Check the SDA/SCL rise time on a scope with the full harness fitted: it must reach 0.3->0.7 VDD within 300 ns at 400 kHz. 7 Shut A2 down cleanly before opening SW1.								C																																																																																											
		D	WIRE LIST <table><tr><th>FROM</th><th>TO</th><th>NET</th><th>WIRE</th></tr><tr><td>BT1 +</td><td>F1 in</td><td>+12V_RAW</td><td>red</td></tr><tr><td>F1 out</td><td>SW1 1</td><td>+12V_FUSED</td><td>red</td></tr><tr><td>SW1 2</td><td>TB1 in</td><td>+12V_SW</td><td>red</td></tr><tr><td>TB1 F2 out</td><td>PS1 IN+</td><td>+12V_PI</td><td>red</td></tr><tr><td>TB1 F3 out</td><td>PS2 IN+</td><td>+12V_ESP</td><td>red</td></tr><tr><td>BT1 -</td><td>TB2</td><td>GND</td><td>black</td></tr><tr><td>TB2</td><td>PS1 IN- , PS2 IN-</td><td>GND</td><td>black</td></tr><tr><td>PS1 OUT+ / OUT-</td><td>A2 USB-C</td><td>+5V_PI</td><td>USB cable</td></tr><tr><td>PS2 OUT</td><td>A1 micro-USB</td><td>+5V_ESP</td><td>USB cable</td></tr><tr><td>A1 3V3</td><td>U1 VIN, U2 VIN</td><td>+3V3</td><td>pink</td></tr><tr><td>A1 5V</td><td>MOD01-16 VIN</td><td>+5V_PAD</td><td>orange</td></tr><tr><td>A1 GND</td><td>U1/U2 GND, MOD01-16 GND, U3 GND</td><td>GND</td><td>black</td></tr><tr><td>A1 IO21</td><td>U1 SDA, U2 SDA, U3 SDA</td><td>SDA_MAIN</td><td>blue</td></tr><tr><td>A1 IO22</td><td>U1 SCL, U2 SCL, U3 SCL</td><td>SCL_MAIN</td><td>yellow</td></tr><tr><td>U1 SD0..SD7</td><td>MOD01..MOD08 SDA</td><td>SDA_A0..A7</td><td>blue</td></tr><tr><td>U1 SC0..SC7</td><td>MOD01..MOD08 SCL</td><td>SCL_A0..A7</td><td>yellow</td></tr><tr><td>U2 SD0..SD7</td><td>MOD09..MOD16 SDA</td><td>SDA_B0..B7</td><td>blue</td></tr><tr><td>U2 SC0..SC7</td><td>MOD09..MOD16 SCL</td><td>SCL_B0..B7</td><td>yellow</td></tr><tr><td>U1 A0, A1, A2</td><td>GND</td><td>ADDR_U1 = 0x70</td><td>-</td></tr><tr><td>U2 A0</td><td>+3V3 (A1, A2 to GND)</td><td>ADDR_U2 = 0x71</td><td>-</td></tr><tr><td>U1 RESET, U2 RESET</td><td>+3V3 through 10 k</td><td>RST_MUX</td><td>-</td></tr></table>								FROM	TO	NET	WIRE	BT1 +	F1 in	+12V_RAW	red	F1 out	SW1 1	+12V_FUSED	red	SW1 2	TB1 in	+12V_SW	red	TB1 F2 out	PS1 IN+	+12V_PI	red	TB1 F3 out	PS2 IN+	+12V_ESP	red	BT1 -	TB2	GND	black	TB2	PS1 IN- , PS2 IN-	GND	black	PS1 OUT+ / OUT-	A2 USB-C	+5V_PI	USB cable	PS2 OUT	A1 micro-USB	+5V_ESP	USB cable	A1 3V3	U1 VIN, U2 VIN	+3V3	pink	A1 5V	MOD01-16 VIN	+5V_PAD	orange	A1 GND	U1/U2 GND, MOD01-16 GND, U3 GND	GND	black	A1 IO21	U1 SDA, U2 SDA, U3 SDA	SDA_MAIN	blue	A1 IO22	U1 SCL, U2 SCL, U3 SCL	SCL_MAIN	yellow	U1 SD0..SD7	MOD01..MOD08 SDA	SDA_A0..A7	blue	U1 SC0..SC7	MOD01..MOD08 SCL	SCL_A0..A7	yellow	U2 SD0..SD7	MOD09..MOD16 SDA	SDA_B0..B7	blue	U2 SC0..SC7	MOD09..MOD16 SCL	SCL_B0..B7	yellow	U1 A0, A1, A2	GND	ADDR_U1 = 0x70	-	U2 A0	+3V3 (A1, A2 to GND)	ADDR_U2 = 0x71	-	U1 RESET, U2 RESET	+3V3 through 10 k	RST_MUX	-	D		
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